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Computer Science Capstone

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The artifact I selected is the same Android inventory management application that I used for my previous enhancements. I originally created the application in CS 360 Mobile Architect and Programming. The application allows users to create an account, log in, and manage inventory items stored in a SQLite database. For this enhancement, I focused on improving the database design and how inventory data is stored and managed.

I included this artifact because the original database was fairly basic and gave me a good opportunity to demonstrate slightly more advanced database concepts. I added an inventory history table that records changes to item quantities, including the previous quantity, new quantity, and a timestamp. I also connected inventory items to the users who created them so that each user can only access and manage their own inventory. These changes created relationships between the several tables. I also used foreign keys and cascading deletion for data integrity. This improved the application by making the database more organized and allowing it to preserve information about inventory changes instead of only storing the current quantity.

This enhancement supports course outcomes three, four, and five. I designed a more structured database solution using relational database principles and updated the application to work with the new design. I used SQLite features such as primary keys, foreign keys, and constraints to manage relationships between the data. The addition of user specific inventory is technically a security improvement, even though it likely wont be too useful on a single device. I believe the enhancement met the outcomes I originally planned, so I do not have any changes to my plans.

This enhancement helped me better understand how relationships between database tables can improve an application, again without many UI changes. Adding a user ID to each inventory item created a one-to-many relationship that allowed multiple users to use the same database while keeping their inventory separate. I also learned more about foreign keys and cascading deletion by connecting inventory items to their history records. Testing the tables with Android Studio's Database Inspector helped me verify that the relationships and history were working correctly.

Below is a checklist of enhancements I made while working on this enhancement:

Expanded Relational Database Structure

- Added an inventory_history table to track changes to inventory quantities.
- Added foreign-key relationships between database tables.
- Established a one-to-many relationship between users and inventory items.
- Established a one-to-many relationship between inventory items and history records.

Implemented User-Specific Inventory

- Added user_id to associate each inventory item with the user who created it.
- Updated item creation to store the authenticated user's ID.
- Updated inventory queries so users only retrieve their own items.
- Ensured inventory updates and deletions only affect the current user's inventory.

Implemented Inventory History Tracking

- Automatically records quantity changes in inventory_history.
- Stores the previous and new quantity for each change.
- Automatically records a timestamp for each history entry.

Improved Data Integrity

- Enabled SQLite foreign-key constraint enforcement.
- Used primary and foreign keys to maintain relationships between records.
- Added ON DELETE CASCADE so related history records are removed when an inventory item is deleted.

Updated Database Version and Migration

- Added User-Scoped Inventory Management
- Added a “Clear Inventory” option for removing all inventory belonging to the current user.
- Ensured clearing one user's inventory does not affect another user's records.
- Added a confirmation dialog before clearing inventory to prevent accidental deletion.

Tested Database Functionality

- Tested inventory using multiple user accounts.
- Verified that users only see and modify their own inventory.
- Verified quantity changes create the correct history records.
- Verified timestamps using Android Studio's Database Inspector.
- Verified foreign-key relationships and cascading deletion.

Reference

Android Developers. (2026, March 6). *Debug your database with the Database Inspector*. https://developer.android.com/studio/inspect/database